BST Physics Curriculum Vol 6 Chapter 11 — Non-Equilibrium Thermodynamics v0.4 (refilled per Cal #104; INTERNAL register Vol 4 Ch 9 cross-link per Cal #50)

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Vol 6 Chapter 11 — Non-Equilibrium Thermodynamics

Chapter motivation

Standard non-equilibrium thermodynamics extends equilibrium theory to systems out of equilibrium (heat flow, chemical reactions in progress, biological systems). Key results:

- Onsager reciprocity (1931 Nobel): cross-coupling coefficient matrix L_ij is symmetric (L_ij = L_ji); generalizes Maxwell's reciprocity
- Fluctuation-dissipation theorem (FDT): equilibrium fluctuations determine non-equilibrium dissipation rates; Callen-Welton 1951
- Entropy production rate: $\sigma = \Sigma$_i J_i X_i (currents × thermodynamic forces); positive for irreversible processes
- Prigogine 1977 Nobel dissipative structures: far-from-equilibrium systems can spontaneously self-organize (chemical oscillations, biological pattern formation)

BST identifies non-equilibrium thermodynamics as substrate cascade evolution between T2417 4-Zone Commitment Cycle phases: - Zone-1 absorption → Zone-2 commitment → Zone-3 emission → Zone-4 outer-edge - Onsager reciprocity from substrate-symmetry constraints - FDT inherits substrate-tick fluctuation structure + K59 cyclotomic-chain irreversibility (Vol 6 Ch 3 second law) - Entropy production rate = substrate-tick Shannon-entropy rate per K59 7-step chain irreversibility

Cosmological cross-link Vol 4 Ch 9 INTERNAL register per Cal #50: substrate annealing toward Λ-saturated state IS cosmological-scale non-equilibrium evolution per Casey-named #7 D_IV^5 Rigidity Principle + #8 SCMP STANDING.

Section 11.0b — Reader-grade 3-level pedagogy

Level 1: Non-equilibrium thermodynamics = substrate cascade evolution between T2417 4-Zone Commitment Cycle phases; Onsager reciprocity from substrate-symmetry constraints; entropy production rate from substrate-tick K59 cyclotomic irreversibility; cosmological-scale cross-link to Vol 4 Ch 9 substrate annealing toward Λ-saturated state (INTERNAL register per Cal #50 DOUBLE-LOCKED EXTERNAL).

Level 2 (graduate-physicist): Standard non-equilibrium: Onsager 1931 reciprocity L_ij = L_ji (cross-coupling matrix symmetric); FDT relating equilibrium fluctuations to dissipation; entropy production σ = Σ_i J_i X_i (current × thermodynamic force) ≥ 0 for irreversible processes. BST substrate-cartography: substrate cascade through T2417 4-Zone (Zone-1 absorption → Zone-2 commitment → Zone-3 emission → Zone-4 outer-edge) describes substrate non-equilibrium evolution per Koons tick. Onsager reciprocity = substrate-symmetry constraint (substrate doesn't care about cross-coupling direction at substrate-tick scale — time-reversal symmetry at substrate level). FDT inherits substrate-tick fluctuation structure: per K59 RATIFIED Tuesday, substrate-tick computation is 7-step cyclotomic projection chain on GF(128); fluctuations at substrate-tick level + dissipation rate at macroscopic level connected via substrate-symmetry. Entropy production rate = substrate-tick Shannon-entropy rate per K59 chain irreversibility. Far-from-equilibrium dissipative structures (Prigogine 1977: chemical oscillations like Belousov-Zhabotinsky reaction, biological pattern formation): substrate-cartography reading as substrate-zone reorganizations driven by external Zone-1 absorption flux. Cosmological cross-link Vol 4 Ch 9 INTERNAL register per Cal #50: substrate annealing toward Λ-saturated state IS cosmological-scale non-equilibrium evolution per Casey-named #7 D_IV^5 Rigidity Principle + #8 SCMP STANDING; per Cal #50 DOUBLE-LOCKED EXTERNAL: external materials use operational language only ("substrate identifies non-equilibrium evolution patterns"), never "universe is annealing toward Λ" metaphysical assertion externally.

Level 3 (5th-grader accessible): Non-equilibrium thermo describes systems out of equilibrium (heat flowing, chemical reactions in progress, biological systems). BST identifies this as substrate cascade evolution through the 4-zone cycle (T2417). Onsager's symmetry rule comes from substrate symmetries. At cosmological scale, non-equilibrium = substrate annealing toward Λ-saturated state per Vol 4 Ch 9 (INTERNAL register per Cal #50 — external materials use operational language only).

Section 11.1 — Standard Non-Equilibrium Thermodynamics

Onsager reciprocity (1931 Nobel): cross-coupling matrix L_ij = L_ji for linear-response coefficients. J_i = Σ_j L_ij X_j with thermodynamic forces X_j. Symmetry from microscopic time-reversal invariance.

Fluctuation-dissipation theorem (Callen-Welton 1951): $\langle \delta A(0)\delta A(t)\rangle \propto \mathrm{Im}[\chi(\omega)]$ — equilibrium two-point fluctuation function determines dissipation susceptibility $\chi(\omega)$. Cornerstone of linear-response theory.

Entropy production rate: $\sigma = \Sigma\_i\ J\_i\ X\_i \geq 0$ for irreversible processes.

Prigogine 1977 Nobel dissipative structures: far-from-equilibrium systems spontaneously self-organize into ordered structures (Belousov-Zhabotinsky chemical oscillations, biological pattern formation, weather patterns).

Section 11.2 — Substrate T2417 4-Zone Cascade

T2417 (Wednesday): substrate operates via 4-zone cycle at per-tick scale: - Zone-1 absorption: substrate absorbs input information (per-tick GF(128)^k input) - Zone-2 commitment: substrate computes per-tick state evolution (Casimir-K-type evolution) - Zone-3 emission: substrate emits per-tick output (residual state → next-tick absorption) - Zone-4 outer-edge: substrate slow-evolving residual integrating to Λ-saturated vacuum state

Non-equilibrium evolution = substrate cascade through these zones per Koons tick.

Section 11.3 — Onsager Reciprocity Substrate-Cartography

Onsager L_ij = L_ji follows from substrate time-reversal symmetry at substrate-tick scale (cross-link T2433 T-reversal substrate operator). Substrate doesn't care about cross-coupling direction at substrate-tick scale.

Section 11.4 — FDT Substrate Substrate-Cartography

FDT $\langle \delta A(0)\delta A(t)\rangle \propto \mathrm{Im}[\chi(\omega)]$ inherits substrate-tick fluctuation structure: per K59 RATIFIED Tuesday, 7-step cyclotomic projection chain on GF(128) generates substrate-tick fluctuations + dissipation. Equilibrium fluctuation amplitude + dissipation rate connected via substrate-symmetry.

Section 11.5 — Entropy Production Substrate-Tick Rate

$\sigma = \Sigma\_i\ J\_i\ X\_i \geq 0$ = substrate-tick Shannon-entropy rate per K59 chain irreversibility (Vol 6 Ch 3 second law).

Macroscopic irreversibility = many-tick integration of substrate-tick projection-irreversibility.

Section 11.6 — Prigogine Dissipative Structures Substrate-Cartography

Far-from-equilibrium dissipative structures (Belousov-Zhabotinsky, biological pattern formation): substrate-zone reorganizations driven by external Zone-1 absorption flux.

Substrate-cartography reading: spontaneous self-organization = substrate-zone-projection equivalence-class reorganization at large Zone-1 flux.

Section 11.7 — Cosmological Cross-Link Vol 4 Ch 9 INTERNAL Register

Cal #50 DOUBLE-LOCKED EXTERNAL: substrate-cognition + cosmology-cognition combined territory; this section's substrate-cycle metaphysical framing is INTERNAL REGISTER ONLY. External materials use operational language only.

Internal-register: substrate annealing toward Λ-saturated state IS cosmological-scale non-equilibrium evolution per Casey-named #7 D_IV^5 Rigidity Principle + #8 SCMP STANDING. Cosmological cycle hypothesis (Vol 4 Ch 9 v0.3) provides substrate-cartography framing.

External-register: "BST identifies cosmological observational signatures testable at future precision surveys (DESI + Euclid + Roman + SKA + LiteBIRD + CMB-S4 + PICO ~2030-2040; Task #267)".

Section 11.8 — Honest scope + falsifiers

Item	Status	Falsifier
Onsager reciprocity substrate-symmetry derivation	Substrate-cartography framework	Counter-example Onsager-violating system at substrate scale
FDT substrate-tick fluctuation framework	K59 RATIFIED foundation	Failure of FDT in measured systems
Entropy production rate from K59 irreversibility	Vol 6 Ch 3 cross-link	Counter-example entropy-decreasing isolated system
Dissipative structures Prigogine substrate-cartography	Substrate-cartography framing	Failure of substrate-zone reorganization model
Cosmological substrate annealing	Vol 4 Ch 9 INTERNAL framing per Cal #50	Task #267 observational signatures (multi-year)

Open scope (multi-week to multi-month): - Quantitative substrate-tick fluctuation derivation from K59 7-step chain - Onsager-reciprocity precision substrate-mechanism via T2433 T-reversal cross-link - Prigogine dissipative-structure substrate-zone-reorganization explicit model

Section 11.9 — Connection

- Vol 6 Ch 3 Entropy + Second Law (K59 cyclotomic irreversibility)

- Vol 4 Ch 9 Cosmological Cycle Hypothesis (INTERNAL register cross-link)
- T2417 4-Zone Commitment Cycle + T2433 T-reversal + Casey-named #7 D_IV^5 Rigidity + #8 SCMP

— Lyra, Vol 6 Ch 11 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT